

Example 1. Let $f(x, y) = \cos x \ln y$ and let $\vec{r}(t)$ be a differentiable path so:

$$\begin{cases} \vec{r}(0) = \langle 0, 1 \rangle \\ \vec{r}'(0) = \langle -1, 1 \rangle \end{cases}$$

Let $g(t) = f(\vec{r}(t))$ and find $g'(0)$.

Solution. Note that $g(t) = f(\vec{r}(t))$ merely tells us to think of $f(x, y)$ as

by setting $\langle x, y \rangle =$

Thus we may think of $g'(t) =$

Applying the path version of the chain rule gives

$$\dots =$$

We evaluate

$$\vec{\nabla} f(x, y) =$$

and now we evaluate at $t = 0$ to obtain

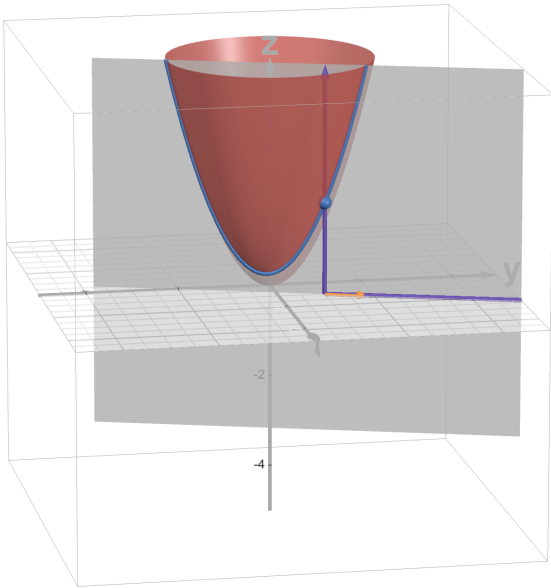
$$\langle x, y \rangle =$$

$$\vec{\nabla} f(x, y) =$$

$$\vec{r}'(0) =$$

$$g'(0) =$$

A. **Directional Derivatives.** We measure slopes in other cross-sections.



Consider graph $z = f(x, y)$ and point with coordinates

$$x = a, y = b, z =$$

We write:

$$\vec{a} =$$

Select a **unit** direction $\vec{u}$.

The slope of the cross-section at P in direction $\vec{u}$ is the **directional derivative**:

We compute it using the chain rule.

Input $\vec{x} = \langle x, y \rangle$ in the cross-section may be parametrized as

$$\vec{x} =$$

By the chain rule, the slope is then

$$\frac{df}{dt} =$$

Evaluating at $\vec{x} = \vec{a}$ yields the directional derivative.

Let f be a differentiable function of $\vec{x}$ and $\vec{a}$ be an input, $\vec{u}$ a **unit** direction.

Then the directional derivative of f in direction $\vec{u}$ may be calculated as

Example 2. Find the rate of change of:

$$f(x, y) = xye^y$$

at $(-2, -1)$ in the unit direction of $\langle -1, -1 \rangle$.

Solution. A rate of change in a particular direction is a directional derivative.

First we replace the direction with a unit direction by

This gives: $\vec{u} =$

Next we calculate the gradient

$$\vec{\nabla} f(x, y) =$$

and substitute in the coordinates $x = -2$ and $y = -1$ of the given point.

$$\vec{\nabla} f(-2, -1) =$$

Lastly we compute the directional derivative as a dot product.

$$D_{\vec{u}} f(-2, -1) =$$

B. Angles and Directional Derivatives. Let θ be the angle between $\vec{u}$ and $\vec{\nabla} f(\vec{a})$.

$$\text{Then: } D_{\vec{u}} f(\vec{a}) =$$

Thus the directional derivative is largest when:

and the directional derivative is smallest when:

and is zero when:

Example 3. An ant is on a hot pan.

Its position is measured by x centimeters right and y centimeters up.

Temperature T in fahrenheit depends differentiably on x and y .

Suppose the ant is at point $(1, 2)$ on the hot pan. If it were to

- moves right, then the temperature would increase at a rate of 12 degrees/cm
- move up, then the temperature would decrease at a rate of 5 degrees/cm

We make a sketch.

(a) Find the direction it should move to most rapidly decrease temperature.

Solution. First we note that the given information encodes gradient.

$$\vec{\nabla}T(1,2) =$$

To minimize rate of change we should pick angle from the gradient

$$\theta =$$

In other words we select a direction:

$$\vec{v} =$$

(b) Find the rate of temperature decrease per centimeter in this direction.

Solution. First we convert $\vec{v}$ to a unit direction.

$$\vec{u} =$$

Next we evaluate the directional derivative.

$$D_{\vec{u}}T(1,2) =$$

Alternatively we could have noted that the minimal rate of change is given by

(c) Which directions could the ant move so temperature stays roughly constant?

Solution. To make the rate of change 0 we should pick angle from the gradient

$$\theta =$$

We use the rule that two vectors orthogonal to $\langle a, b \rangle$ are:

to select the following directions orthogonal to $\vec{\nabla}T(1,2) =$